Report on LROM

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Learned Reduced-Operator Models: Short Summary

Goal. We developed and tested a reduced-coordinate modeling strategy for high-dimensional physics calculations. The motivating example was neutron scattering from nuclei using ROSE [1] as a high-fidelity solver and as a reference reduced-basis emulator. The ROSE paper is the main reference for some of the scattering notation used here. The broader idea is not tied to scattering: given a full-order model (FOM) that produces a high-dimensional state $\phi(\alpha)$ for control parameters α, we first construct a lower-dimensional representation

$$\phi(\alpha) \approx \hat{\phi}(\alpha),$$

where the "hat" denotes the chosen approximation. This reduced representation need not be linear. In the present scattering tests, however, we used a reduced-basis expansion

$$\hat{\phi}(\alpha) = \phi_0 + \sum_{i=1}^{n} a_i(\alpha)\phi_i.$$

The central modeling problem is then to learn how the reduced coordinates $a_i(\alpha)$ evolve with α. The central question is whether a learned reduced-operator model (LROM) can reproduce the best reduced-space approximation to the FOM, and in some regimes improve on a standard Galerkin/EIM reduced-order model (ROM) [2].

Core method. For a training set of FOM solutions, we build a reduced basis $\Phi = [\phi_1, \phi_2, \ldots]$ by applying a singular value decomposition to a set of solution snapshots. The motivation for the implicit-equation formalism comes from what a standard Galerkin ROM would produce in a simple case. Suppose, for example, that the only parameter being varied is the real Woods-Saxon depth V_v, while the radial shape of the potential is fixed:

$$V(r; V_v) = -V_v f_{\mathrm{WS}}(r).$$

If the wavefunction is represented using two reduced-basis functions,

$$\hat{\phi}(V_v) = \phi_0 + a_1(V_v)\phi_1 + a_2(V_v)\phi_2,$$

then projecting the Schrodinger equation onto the reduced space gives a small implicit system for the two reduced coefficients. Writing $a(V_v) = (a_1(V_v), a_2(V_v))^T$, the projected reduced equation has the form

$$(M_0 + V_v M_1)\, a(V_v) = b_0 + V_v b_1.$$

Here M_0 and M_1 are 2×2 reduced matrices and b_0 and b_1 are 2×1 reduced source vectors. In a traditional Galerkin ROM, these objects are computed by explicitly projecting the high-dimensional operators, such as the kinetic and potential terms, onto the reduced basis. The LROM idea is to keep this same implicit reduced-equation structure, but learn the reduced matrices and vectors from FOM data instead:

$$\left(\widetilde{M}_0 + V_v \widetilde{M}_1\right) a = \widetilde{b}_0 + V_v \widetilde{b}_1.$$

More generally, when there is a list of controlling parameters α, we write

$$M(p(\alpha))\, a = b(p(\alpha)),$$

where $p(\alpha)$ are physically motivated predictors encoding how the underlying operator changes with the control parameters, and the entries of the reduced operator blocks are trainable.

To train the model, we first compute target reduced coordinates $a_{\mathrm{LS}}(\alpha)$ by least-squares projection of each FOM state onto the chosen reduced representation. A key discovery is that, whenever M and b are linear in the trainable operator blocks, training can be done by evaluating the proposed equation on the known target coefficients a_{LS}:

$$M(p(\alpha_i))\, a_{\mathrm{LS}}(\alpha_i) - b(p(\alpha_i)) \approx 0.$$

For the simple V_v-only example, this means fitting the entries of $\widetilde{M}_0$, $\widetilde{M}_1$, $\widetilde{b}_0$, and $\widetilde{b}_1$ so that

$$\left(\widetilde{M}_0 + V_{v,i}\widetilde{M}_1\right) a_{\mathrm{LS}}(V_{v,i}) - \left(\widetilde{b}_0 + V_{v,i}\widetilde{b}_1\right) \approx 0$$

for all training values $V_{v,i}$. Since this residual is linear in the unknown matrix and vector entries, the training problem becomes a one-shot linear least-squares solve. This is the residual-fit LROM (RF-LROM) viewpoint.

This should be contrasted with an older coefficient-fit LROM (CF-LROM), where one solves the current implicit system during training to obtain $a_{\mathrm{pred}}(\alpha)$, and then minimizes

$$\|a_{\mathrm{pred}}(\alpha) - a_{\mathrm{LS}}(\alpha)\|^2.$$

The coefficient-fit approach is more expensive because each optimization step requires repeatedly solving reduced implicit systems. The residual-fit approach moves the solve out of the training loop: the known FOM-projected coordinates are inserted directly into the proposed equation, exposing a linear problem for the unknown reduced operators.

Main discoveries. Below we list the major discoveries we have made so far in the specific scattering example.

Speed. The RF-LROM formulation is dramatically faster to train than CF-LROM. In a simple one-parameter test, we obtained the optimal reduced equations in about 5×10^{-4} seconds using the residual-fit linear solve, compared with roughly 20 seconds for the nonlinear coefficient-fit training. The key reason is that RF-LROM does not solve the reduced implicit equation during training; it evaluates the proposed equation directly on the FOM-projected target coordinates.

Identifiability and gauge freedom. The residual-fit formulation also exposed an important identifiability issue. For simple one-parameter tests, the residual design matrix had a nontrivial nullspace: many different reduced operators reproduced the same solution manifold. This explained why unconstrained residual fits could yield ill-conditioned reduced systems. Small regularization terms or gauge choices selected stable representatives from this equivalence class. We then adopted a formulation that focuses on constructing the LROM around a central value of the controlling parameters, α_c. With this choice, we set the reference state to the central scattering solution, $\phi_0 = \phi(\alpha_c)$, and fix the constant matrix to $M_0 = I$. This removes a large left-multiplication redundancy:

$$\left[I + \sum_{j=1}^{K} p_j(\alpha) M_j\right] a = \sum_{j=1}^{K} p_j(\alpha) b_j.$$

Here the predictors are defined so that $p_j(\alpha_c) = 0$, ensuring that the central solution has $a(\alpha_c) = 0$. In the numerical experiments performed so far, this choice also removed the nullspace of the residual design matrix.

Predictors matter. The simple V_v-only example is special because changing the depth only rescales a fixed radial shape. In that case an affine equation such as

$$(I + \Delta V_v M_V)\, a = \Delta V_v b_V$$

is a natural local model, where $\Delta V_v = V_v - V_{v,c}$. If we also vary the radius R_v, a first extension is to add another linear term,

$$(I + \Delta V_v M_V + \Delta R_v M_R)\, a = \Delta V_v b_V + \Delta R_v b_R.$$

This worked reasonably well for small parameter variations. However, it deteriorated over wider boxes because changing R_v does not simply add a fixed operator direction: it shifts the Woods-Saxon surface in radius. Thus the operator dependence is not well described, over large ranges, by independent linear terms in the raw parameters.

To address this, we replaced raw-parameter predictors with operator-informed predictors. Instead of using only quantities such as ΔV_v, ΔR_v, or ΔE, we evaluate the scaled optical potential itself at selected points. In its simplest form, the predictor associated with the point s_j is

$$p_j(\alpha) = U(s_j;\alpha) - U(s_j;\alpha_c),$$

where $U(s;\alpha)$ is the scaled potential entering the Schrodinger equation and α_c is the central parameter point. The LROM equation then becomes

$$\left(I + \sum_{j=1}^{K} p_j(\alpha) M_j \right) a = \sum_{j=1}^{K} p_j(\alpha) b_j.$$

This keeps the training problem linear in the unknown reduced matrices and vectors, but lets the dependence on the original physical parameters be nonlinear through the predictors $p_j(\alpha)$. In the implementation we optionally normalize the predictor variations by their typical training-set magnitudes for numerical conditioning; this does not change the structure of the method, only the scaling of the fitted matrices and vectors.

The points s_j were chosen using a delta-maxvol procedure. We first formed the operator variations around the central potential,

$$\Delta U(s;\alpha_i) = U(s;\alpha_i) - U(s;\alpha_c),$$

over the training set. We then computed a low-rank representation of these variations and selected spatial points s_j that were maximally informative for reconstructing them. In this sense the predictors are not arbitrary samples of the potential: they are collocation-like measurements chosen to capture the dominant ways in which the operator changes over the training region. This made the LROM depend on V_v, R_v, a_v, and E through the same nonlinear operator features seen by the Schrodinger equation, and was much more robust than using separable linear predictors in the raw parameters alone.

Although this maxvol procedure uses information about the full potential, which may not be directly available in more complex models, the broader lesson is that good predictors should be inspired by the structure of the underlying equations. In future versions, predictor locations or predictor parameters could themselves be treated as higher-level hyperparameters. One could then use a hierarchical optimization strategy: for a fixed choice of predictor hyperparameters, solve the linear residual-fit problem for the reduced matrices in one shot; then adjust the predictor hyperparameters to improve validation error or conditioning. This would preserve the main computational advantage of RF-LROM while allowing the predictor family to adapt to the problem.

The basis floor matters. Throughout the tests, we compared LROM emulator errors with the least-squares projection floor obtained by fitting the FOM wavefunction directly in the chosen reduced representation. This distinction is essential. If the LS floor is large, meaning that the reduced coordinates are already at their limit in reproducing the high-dimensional object, improving the reduced equation cannot fix the problem; the reduced representation itself is insufficient. If the LS floor is small but the LROM or ROM error is large, then the limitation lies in the learned or projected reduced equations.

Representative results. For cross sections with fixed energy and variation of the 10 Woods-Saxon parameters varied around 30% of their central values as the ROSE paper, the potential-predictor LROM achieved accurate and fast online evaluations. The main comparison was made using computational accuracy versus time (CAT) plots, where each point corresponds to one test parameter sample. The vertical axis is the median pointwise relative error in the differential cross section, while the horizontal axis is the online evaluation time per sample. In the optimized packed implementation, increasing the number of predictor operators K improved accuracy at almost no speed penalty. For example, in one saved CAT study the median pointwise cross-section errors were approximately

method	median time/sample [s]	median pointwise error
ROSE $(4,8)$	5.2×10^{-4}	2.9×10^{-2}
ROSE $(10,10)$	5.9×10^{-4}	2.8×10^{-3}
LROM $K = 9$	1.8×10^{-4}	2.5×10^{-3}
LROM $K = 12$	2.0×10^{-4}	8.4×10^{-4}

for the same test cloud. The comparison is shown in Fig. 2. For these configurations, the LROM reaches ROSE-like accuracy with a cheaper online stage, and in some cases improves on ROSE at comparable or lower cost.

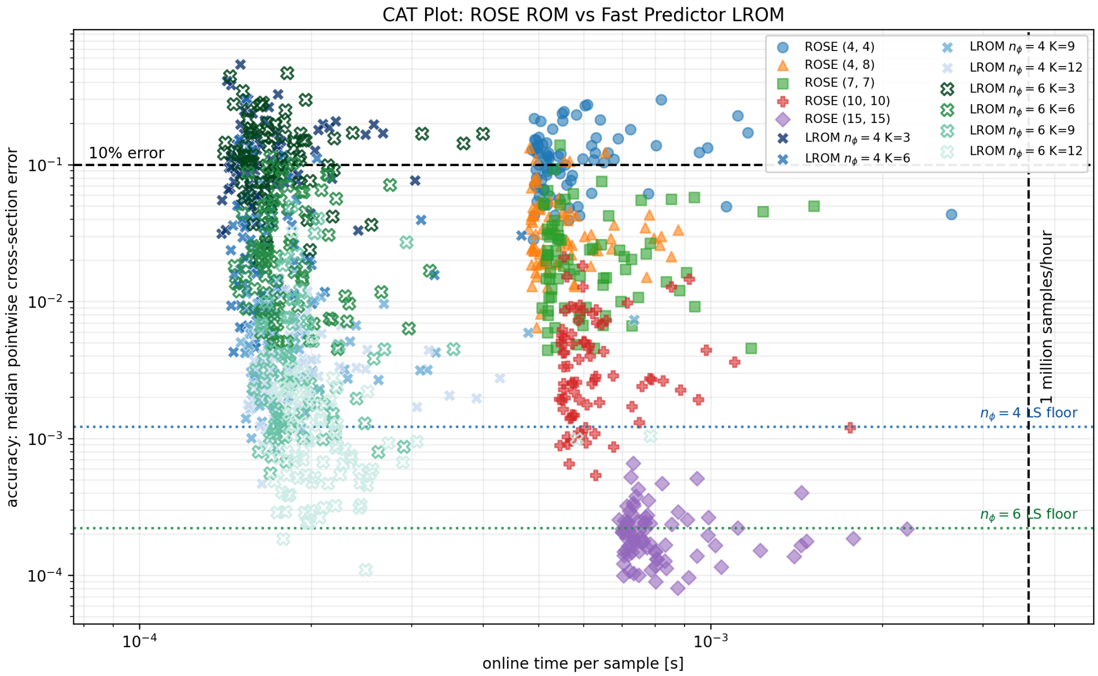

Figure 1: CAT comparison between ROSE ROM basis configurations and fast packed predictor LROMs.

n	K	eqs/ch.	params/ch.	params/21 ch.
4	3	4	60	1260
4	9	4	180	3780
4	12	4	240	5040
6	3	6	126	2646
6	9	6	378	7938
6	12	6	504	10584

Table 1: Representative LROM reduced-system sizes used in CAT-plot cross-section calculations.

Equations and parameters. For these cross-section tests, the LROM is trained separately for each partial-wave channel. If a channel uses n reduced-basis coefficients and K predictor operators, then the online equation for that channel has n complex equations for n unknown coefficients. The learned operator contains K complex matrices $M_j \in C^{n\times n}$ and K complex vectors $b_j \in C^n$. Thus the number of learned complex parameters per channel is

$$N_{\text{par}}^{\text{channel}} = K(n^2 + n).$$

For a cross-section calculation with partial waves up to $L_{\max} = 10$, there is one channel for $\ell = 0$ and two spin-orbit channels for each $\ell = 1, \ldots, 10$, giving 21 channels total. Representative sizes used in the CAT-plot calculations are shown in Table 1.

These counts are for complex parameters. The fits are performed separately for each partial-wave channel, not as one coupled system across all channels. If the training set has N_{train} FOM samples, then each channel provides

$$N_{\text{eq}}^{\text{channel}} = N_{\text{train}} n$$

complex residual equations for the residual-fit linear system. For example, if $N_{\text{train}} = 100$, the $n = 6$, $K = 12$ case has 600 complex residual equations for 504 complex unknowns per channel. Across all 21 channels, this corresponds to 12600 complex residual equations and 10584 complex unknowns. If one stores real and imaginary parts separately, both the equation count and the parameter count double.

Next direction: fitting data. The long-term goal is to use the LROM not only as an emulator of an optical potential model, but as a structured model class that can be fine-tuned to experimental data. The natural

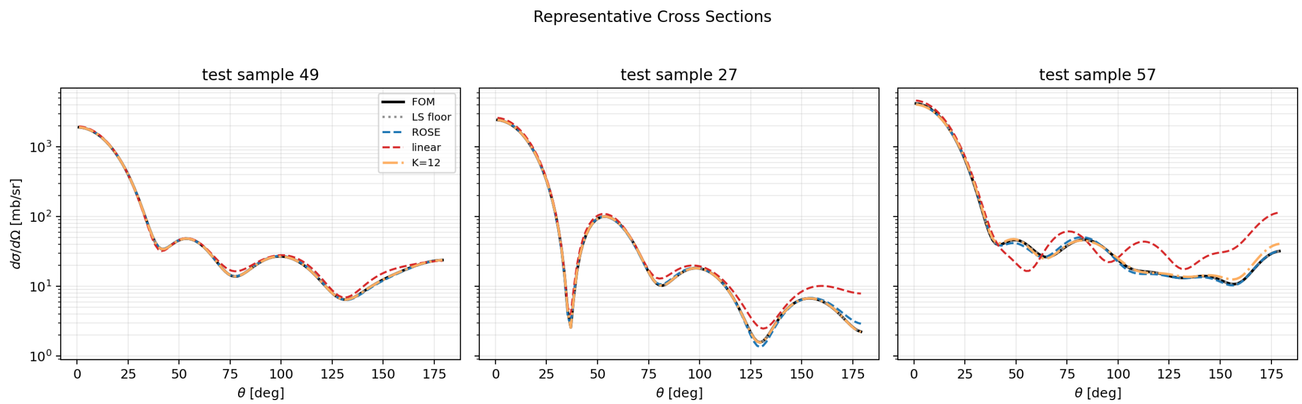

Figure 2: Representative cross sections for some test parameters. "linear" curve is using linear form of the operator equations instead of predictors.

workflow is: train an LROM on FOM-generated states to learn a stable physics-informed operator family, then adjust selected operator blocks or predictor combinations using observable data, such as cross sections. This would explore a broader model space than traditional optical potential parameter fitting while retaining reduced-equation structure.

Several appendices with details follow

A Current Best LROM Workflow for Cross Sections

This appendix summarizes the current version of the LROM workflow used in the most successful cross-section tests. The description is specific to the scattering implementation, but the same logic should transfer to other FOMs.

A.1 Training data and central state

Choose a central parameter point α_c. In the scattering calculations, α may contain Woods-Saxon parameters, isotope information, and energy. For each partial wave and channel, compute the central FOM wavefunction

$$\phi_c = \phi(\alpha_c),$$

and set

$$\phi_0 = \phi_c.$$

Then generate FOM wavefunctions $\phi(\alpha_i)$ over a training set. The reduced basis is built from centered snapshots,

$$\Delta\phi(\alpha_i) = \phi(\alpha_i) - \phi_c,$$

using an SVD. The LROM wavefunction approximation is therefore

$$\hat{\phi}(\alpha) = \phi_c + \Phi a(\alpha).$$

At the central point we want $a(\alpha_c) = 0$, so that $\hat{\phi}(\alpha_c) = \phi_c$ exactly.

A.2 Least-squares target coordinates

For each training wavefunction, compute the least-squares reduced coordinates

$$a_{\mathrm{LS}}(\alpha_i) = \arg\min_a \|\phi(\alpha_i) - \phi_c - \Phi a\|_2.$$

These are the target coordinates for training the reduced equation. They also define the LS floor: the best wavefunction accuracy possible with the chosen reduced representation.

A.3 Predictor construction

For each partial wave/channel, define the scaled potential $U(s;\alpha)$ on the ROSE operator coordinate $s = kr$. We form potential variations around the central parameter,

$$\Delta U(s;\alpha_i) = U(s;\alpha_i) - U(s;\alpha_c),$$

and use a low-rank decomposition plus delta-maxvol selection to choose collocation-like points s_j. The raw predictors are then

$$p_j(\alpha) = U(s_j;\alpha) - U(s_j;\alpha_c).$$

In the code, these predictors may be normalized by a typical training-set magnitude for numerical conditioning. This normalization is optional in the mathematical structure because it can be absorbed into the fitted matrices and vectors.

A.4 Central-gauge LROM equation

The current preferred LROM ansatz fixes the constant matrix to the identity:

$$\left[I + \sum_{j=1}^{K} p_j(\alpha) M_j\right] a(\alpha) = \sum_{j=1}^{K} p_j(\alpha) b_j.$$

Because $p_j(\alpha_c) = 0$, the equation at the central point becomes

$$I a(\alpha_c) = 0,$$

so $a(\alpha_c) = 0$ and the central FOM wavefunction is reproduced exactly.

A.5 Residual-fit training

The trainable objects are the matrices M_j and vectors b_j. For each training point, insert the least-squares coordinate $a_{\mathrm{LS}}(\alpha_i)$ into the proposed equation and form the residual

$$r_i = \left[I + \sum_{j=1}^{K} p_j(\alpha_i) M_j \right] a_{\mathrm{LS}}(\alpha_i) - \sum_{j=1}^{K} p_j(\alpha_i) b_j.$$

Since the residual is linear in the unknown entries of M_j and b_j, all operator blocks are obtained from one linear least-squares solve. No reduced systems need to be solved during training.

A.6 Online prediction and cross sections

For a new parameter value α_*, the online stage is:

1. Evaluate the predictors $p_j(\alpha_*)$ from the scaled potential at the selected points s_j.
2. Assemble the reduced matrix and right-hand side,

$$M_*(\alpha_*) = I + \sum_{j=1}^{K} p_j(\alpha_*) M_j, \qquad b_*(\alpha_*) = \sum_{j=1}^{K} p_j(\alpha_*) b_j.$$

3. Solve the small reduced system

$$M_*(\alpha_*) a(\alpha_*) = b_*(\alpha_*).$$

4. Reconstruct each partial-wave/channel wavefunction,

$$\hat{\phi}(\alpha_*) = \phi_c + \Phi a(\alpha_*).$$

5. Extract the scattering information from the reconstructed wavefunctions and combine the partial waves to produce the differential cross section.

Everything that does not depend on the new parameter value, such as reduced bases, fitted matrices, selected predictor points, and some asymptotic quantities, can be precomputed and packed. The online time should include only the parameter-dependent steps: predictor evaluation, reduced-system assembly and solve, wavefunction reconstruction, and observable evaluation.

References

[1] ROSE: A reduced-order scattering emulator, `https://arxiv.org/abs/2312.12426`.

[2] Reduced-basis and empirical interpolation notes for scattering, `https://dr.ascsn.net/scattering/eim.html`.